Automated Selective Color-wise Colony Counting on Spread Plates using Deep Learning and Human-in-the-loop Integration

Abstract—**Microbial colony counting on the spread plate is one of the most important processes in microbiology lab. This lets the researchers understand microbial growth. In general, the researchers must count manually, which is prone to human error and is time-consuming. Especially for the plate that has more than one species growth which creates higher complication for the person who performs this task. This research proposes an automatic colony counting approach via two working modes, including Direct Species Counting Mode, which count the five representative strains directly using YOLOv8, and Selective Color-wise Counting Mode, which count the microbial based on their color using YOLOv8 integrated with color space conversion and K-means techniques to enhance the color analysis. Furthermore, for practical use, the researchers also develop Human-in-the-loop web applications workflow that allows the user to correct the result using manual method. However, from the experiments using the automation, both modes can work well at 87.8 percent accuracy for colony detection, and 89.1 percent accuracy for color analysis, in Selective Color-wise Counting Mode. The collaboration between automation and the addition of Human-in-the-loop functionality significantly reduces workload and resolves ambiguity in color separation for biological analysis.**

Keywords—Microbial Colony Counting, Deep Learning, Object Detection, Clustering, Color Space, Human-in-the-loop, Digital Image Processing

I. Introduction

Colony counting on petri dishes is a fundamental task in microbiology for assessing microbial growth [1]. In practice, this process is typically performed manually, which is time-consuming and prone to human error [2][3]. One key challenge arises from the need to distinguish colonies based on visual characteristics, particularly color, which is often used to separate different microbial groups on the same plate. However, when colonies exhibit similar color tones or when plates are densely populated, accurate counting becomes difficult and inconsistent. These challenges motivate the need for automated approaches that can reliably interpret both colony morphology and color information.

To address these issues, image processing technologies have been introduced to assist in colony counting, many works use thresholding, edge detection, and morphological techniques to detect the microbial shape. However, this traditional method has limitations in practical applications [2]. Firstly, performance is significantly degraded when faced with images taken with low environment-control, for example, the images taken by the mobile phone cameras or images with uneven lighting. Secondly, there are limitations in counting colonies by color. Most tools focus only on total counting but cannot effectively differentiate between color groups to distinguish different types of microorganisms in the same plate [4].

Furthermore, in terms of datasets for the model training, such as AGAR, even if there are many sample images, it still lack of color-coded colony labels. This makes it impossible to directly train models for color classification.

With the growth of AI and the deep learning technique, we can develop an AI-based web application for selective color-wise counting. This can combine two main techniques, including the YOLOv8 model to detect and count the colonies in the petri-dish (spread plate), and K-means [5] Clustering (unsupervised machine learning) to cluster the colonies using color shade. This research proposes a solution using Color Manipulation techniques in color spaces like HSV. This method simulates a variety of colony colors while keeping the original textures and reduces the effect of the uneven light problem. Furthermore, if the microbial sample category is available in the AGAR dataset, our work can classify the microbial colonies using supervised machine learning to label microbial colonies.

The system is designed as a web application that includes a Human-in-the-loop approach. Users can upload images, see results, and use tools to manually edit the data, such as adding, deleting, changing color groups, or using the undo function. This system serves as a helpful tool that significantly reduces working time and minimizes human errors in the laboratory.

In summary, the main contributions of this research are

1. The color manipulation technique for colony color analysis.
2. Machine learning integration using YOLOv8 for localization and K-means for color clustering to make selective color-wise counting for unknown species microbial.
3. A Human-in-the-loop web application, users can interact with the result by using the tools to manually edit the data. This is a practical tool that significantly reduces working time and minimizes human error. In other words, users do not have to start counting from zero and users can know the mistakes created by system.

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II. Background and Related Work

A. Color of microbial

Colony color is an important phenotypic characteristic in microbiology and is commonly used as a preliminary indicator to differentiate microbial species. The observed color may result from microbial pigment production or biochemical reactions on selective and differential media. From a computational perspective, color serves as a useful feature for grouping colonies, particularly when species labels are unavailable. However, its reliability is affected by factors such as lighting conditions, imaging devices, and media variations, which can alter color perception. Therefore, effective use of color in automated systems requires appropriate processing techniques, such as color space transformation, to reduce illumination effects and enable more robust Selective Color-wise grouping and counting.

B. OpenCFU (Related Software)

Nowadays, there are several development of colonies counting tools, especially OpenCFU [2]. The OpenCFU application is the open-source software that use the traditional digital image processing techniques, e.g. thresholding, edge detection, morphological operations, Hough transform, etc. to detect the circle or oval in the image. Although OpenCFU is fast and support the batch processing, it still have the limitation with the image taken without or less environment control, such as the image taken by smartphone or uneven light image. Not only that, this software does not support Selective Color-wise colonies count. If the unwanted type of microbial growth is present in the plate, OpenCFU will count all existed colonies, not only the desired ones.

C. Deep Learning

Deep learning is one of the powerful technique in machine learning field. It was inspired by the neural network structure of the human brain [6]. The main idea of deep learning is the use of multiple layers of mathematics calculation to automatically process complex and large data. The deep learning model can learn to extract the important features from raw data such as images, audio, or text without adjusting by human and use them as a source of decision. This results in high accuracy in computer vision, natural language processing, and deep prediction. These are the underlying technologies for facial recognition systems, self-driving cars, and large-scale language models today [6][7].

D. Color Space Conversion

In this work, we are focusing on the HSV color space conversion to convert the original basic RGB system to HSV. The reason that HSV is better than original RGB is because HSV space stores the color using hue and brightness separation, unlike RGB which store color as intensity. HSV allows us to get the color shades of objects without affecting lighting details. To convert from RGB to HSV color space, it can be calculated using (1), (2), and (3):

$$H = \begin{cases} undefined & ,if\ R' = G' = B' \\ 60° \times \left(\frac{G' - B'}{\Delta}\ mod\ 6\right) & ,if\ \max(R', G', B') = R' \\ 60° \times \left(\frac{B' - R'}{\Delta} + 2\right) & ,if\ \max(R', G', B') = G' \\ 60° \times \left(\frac{R' - G'}{\Delta} + 4\right) & ,if\ \max(R', G', B') = B' \end{cases} \quad (1)$$

$$S = \begin{cases} 0 & ,\max(R', G', B') = 0 \\ \frac{\Delta}{\max(R', G', B')} & ,otherwise \end{cases} \quad (2)$$

$$V = \max(R', G', B') \quad (3)$$

where R', G', B' are RGB values normalized to closed range $[0, 1]$, and $\Delta = \max(R', G', B') - \min(R', G', B')$ [7][8]. The color space comparison between RGB and HSV can be shown in Fig.1

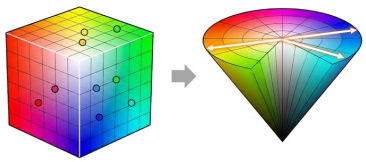


Fig. 1. Comparison between RGB color cube (left) and HSV color cone (right), illustrating the separation of hue (glowing arrow), saturation, and value.

E. Object Detection

Object detection is a group of techniques that combined the ability to classify the image and object localization. This algorithm not only results in category of objects but also knows the area of objects in the image as the group of coordinate numbers. This allows us to draw the bounding box around objects. In this work, we use YOLOv8 [9] as the main model to do object detection. Using YOLOv8, it can show the confidence score of every prediction, so we can know the count of the interested object and can set the level of sensitivity through the setting of the threshold of confidence score.

F. Image Segmentation

Image segmentation is a technique in computer vision field [4][10]. This technique is finer than the common object detection technique. Instead of the bounding box, image segmentation can find the boundary or edge of the object at pixel-level [10]. That let us get the mask of object in the image. By this result, we can calculate area, shape or specific features that we want which exactly existed in the object. Especially, in terms of microbial image processing, this will give us a lot of benefits, since the colonies shape is irregularly oriented or separate adjacent colonies.

G. K-means Clustering

K-means clustering is an unsupervised learning algorithm used to cluster the data based on their similarities. The key idea of this algorithm is to set the number of groups that we want to cluster, then use the random centroids as the initial center of each group. Next, the system will calculate the distance between each

data point and centroids and set the nearest centroid to be the center of its cluster. The coordinate of centroid will be updated using the mean values of the data point of its group. This process must finish using many iterations. In this work, this technique can be applied to cluster the color shades of colonies on the plate, since we do not have prior knowledge about the colors that will appear in the plate.

III. Materials and Datasets

A. Hardware and Software Environment

We use a computer capable of handling image processing to train and inference. The hardware specifications include an Intel Core i5-11400H CPU, 8 GB of RAM, and an NVIDIA GeForce RTX 3050Ti GPU for accelerating model learning.

The software was mainly developed using Python. The core for detection and counting uses the YOLOv8 model. The web application structure is divided into two parts, including (1) the backend, developed with the Flask API for image processing, and (2) the front-end, developed with React for creating a real-time user interface.

B. Datasets

To train and develop machine learning and deep learning models, the petri dish images that are used in this work were obtained from two sources.

- **Public dataset.** The primary source of this research was the Annotated Germs for Automated Recognition (AGAR) dataset [3]. AGAR is an open-source large-scale plate microbial colony image library. The dataset contains over 18,000 high-resolution images with over 336,000 identified colonies across five representative strains, including S. aureus, B. subtilis, P. aeruginosa, E. coli, and C. albicans. The images in this dataset were taken under a variety of environments, including varying light and colony densities to mimic the practical situations. In this work, the AGAR dataset served as the basis for evaluating the performance of proposed color manipulation and human-in-the-loop systems, particularly in addressing the lack of a multi-strain dataset through color change simulation.

- **Local dataset.** The local image dataset was kindly provided by (1) the Microbiology Laboratory, Department of Microbiology, Faculty of Science, Silpakorn University, and (2) the Food Industry Laboratory, King Mongkut's Institute of Technology Ladkrabang. At the time of the study, this dataset contained a total of 220 images. This dataset has no label attached. To prove the predicted number of colonies is correct, we ask two experts in microbiology to help us image by image. To avoid the bias resulting, we blind the predicted result from experts.

C. Datasets Preparation

Although the original AGAR dataset contains over 18,000 images, the researchers did not utilize the entire collection. Instead, the data underwent a rigorous screening process to ensure it matched the specific objectives of this project. The selection was based on the following primary criteria:

- **Selection of Plate and Cultivation Methods**. This study is strictly limited to clear Petri dishes using the spread plate technique. This method was chosen because it produces well-isolated colonies that grow separately. Such distinct features are essential for the model to accurately perform object detection, counting, and Selective Color-wise classification.

- **Exclusion of Low-Quality and Incompatible Data**. A significant portion of the AGAR dataset, around 30%, consists of images that are unsuitable for this model's training. Specifically, we omitted samples labeled as "Uncountable" and "Empty" plates. These categories do not support the model's learning process for counting or species differentiation.

IV. Methods

This work is designed as a web application to allow users access and practical in laboratory. We divide the development into two parts, including model development, and system architecture workflow.

A. Model Development

Making this system can memorize and count the colonies by color cluster (Selective Color-Wise Counting), we must use the image segmentation to segment the group of pixels that have close properties, e.g. color, intensity, texture, etc. This let us understand the edge of the colonies area and extract the real color of each one. We applied the image detection technique named YOLOv8 (You Only Look Once version 8) that has pros in the inference time usage by using all image analysis in one time. This can make the bounding box around the detected area accurately. In this work, we separated the process into two modes, including direct species counting mode and Selective Color-wise Counting Mode.

1) Direct Species Counting Mode

This mode uses the labeled, 5-species, information from AGAR as the source only. We separate the dataset into training set, validation set and test set using 70:10:10 ratio. Then perform the image augmentation including rotating, scaling, noise adding, brightness adjusting and shearing on training set. The training set is used to train YOLO model and validate the performance using validation set. Due to this work having two different YOLO models for two modes, individually, to reduce the confusion, we will call this trained model as YOLO(1). This YOLO(1) has ability to detect and classify the species of each colony. However, this model works well only for 5 well-known species microbial which are available in AGAR dataset. If the species in the plate are other species or unknown another mode is required. The overall pipeline of this YOLO(1) model development is show in Fig. 2 (left)

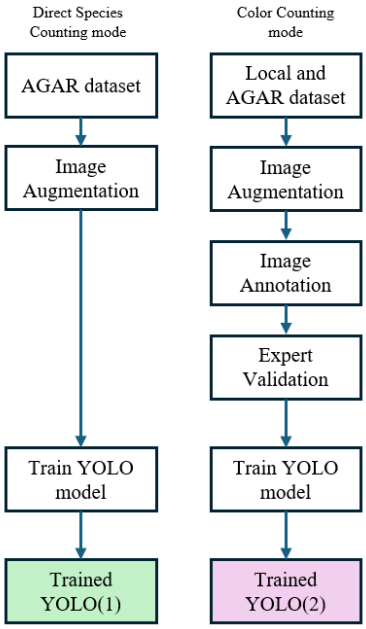


Fig. 2. Overall model development pipeline. Two YOLO model were trained using different methods due to their limiation. Left: YOLO(1) will have ability to specify the species of each colony but the species can classify is limited for 5. Right: YOLO(2) will have more flexibility but cannot specify the exact name of any species.

2) Selective Color-wise Counting Mode

This mode used image data from our local dataset to train the YOLO model. Developing the Selective Color-wise Counting Mode requires a specialized data preparation workflow, as current local dataset does not provide specific color labels. To overcome this limitation, the research team designed the following procedure:

- Every image was annotated using bounding boxes on the Roboflow platform [11]. The system focused solely on identifying the coordinates of general colonies without assigning specific species names to the markers.
- To maintain high data integrity, all annotated images were reviewed and verified by experts before moving to the next phase of development.
- To improve the model's ability to generalize and handle various image conditions, several data augmentation techniques were applied. These included horizontal and vertical flipping, rotations, random rotations, noise addition, and shearing.

After we finished all procedures, we split the data into training set, validation set and test set. Because the dataset is limited and the difficulty of non-labeled dataset evaluation, we reserve the test set for 20 images, and the remaining will be split into 90:10 for training set and validation set. The training set was used to train the YOLO model; we will call this model YOLO(2) for the same reason as YOLO(1). By YOLO(2), we are focusing on object detection to localize the position of every colony on the plate. We do not care about the color on this model. The overall pipeline of this YOLO(2) model development is show in Fig. 2 (right)

B. System Architecture Workflow

The plate image analysis process is designed to integrate deep learning and unsupervised machine learning techniques using a structured data flow as follows:

- Detection and Localization phase.
- In-depth Analysis phase.
- Manual Correction phase.

The overall processing pipeline can be shown as Fig. 3. The information for each block can be explained as follows:

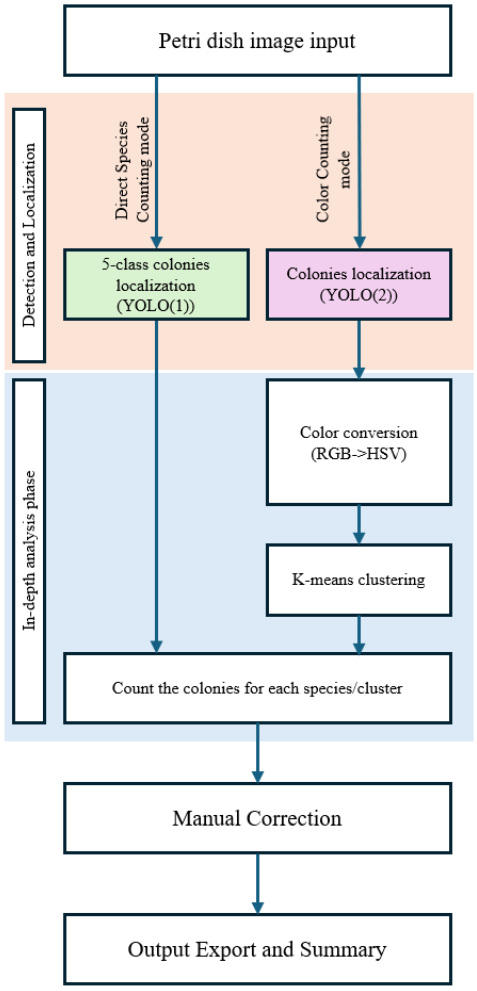


Fig. 3. Overall inference processing pipeline.

1) Detection and Localization

When a user uploads a petri dish image and selects the Direct Species Counting Mode, the system initiates the analysis using a YOLO(1). The system scans the entire plate to detect the location of each colony and analyze its specific features for accurate species identification.

On the other hand, if user selects the Selective Color-wise Counting Mode, another YOLO(2) model which trained on the non-species-label dataset will be activated. By this mode, YOLO(2) will only specify the area of each colony.

2) Color Conversion

Since the color of each species is very close to each other, the shadows and uneven light tend to affect the color. This is

very hard to use the centroid-based clustering method to separate the colors. From this reason, we try to reduce the effect of the light intensity by converting the image to HSV color space. Because HSV color space stores the color and light intensity separately, it allows us to get information of color clearer.

3) Color Clustering

For Selective Color-wise Counting Mode, we average Hue value, in HSV color space, within each colony to represent each colony color. Then we use the K-means algorithm to cluster the color identity using average Hue value of each colony into k groups (depending on user settings). This approach effectively groups colonies with similar color identities into the same cluster, enabling precise selective color-wise counting.

4) Count the Colonies for Each Species/Cluster

Both direct species counting mode and Selective Color-wise Counting Mode, they all give us the label of each colony. The system can count the number of colonies for each label and return all count as the list of results.

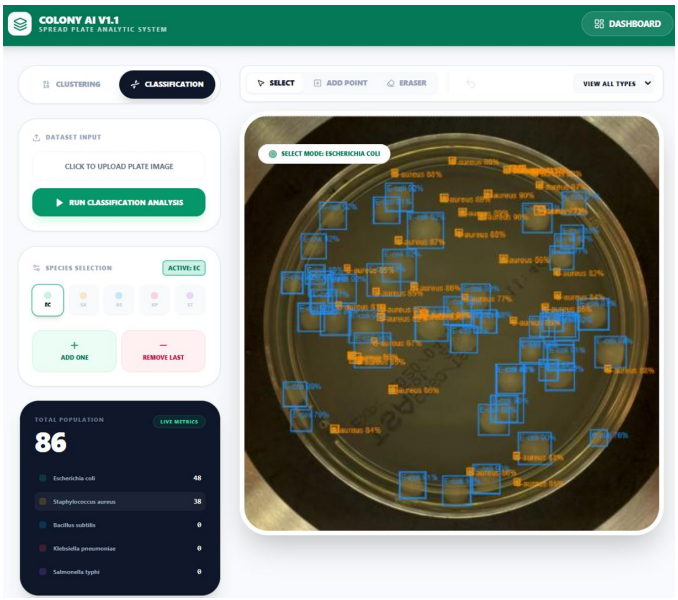


Fig. 4. A user interface that user can correct the mistake from the model using Add mode or Eraser mode. User can also see the counted colonies and specify the correction.

5) Manual Correction

After the automated detection and classification process is finished, the system will display all results on the user interface. In practice, analyzing biological images is difficult because of factors like overlapping colonies, media color changes, or poor lighting. These issues can lead to errors in any predictions. To solve this problem, the system uses a "Human-in-the-loop" method, which allows users to manually check and improve the YOLO or K-means results.

Users can use several interactive tools to manage the data. For example, the 'Add Mode' allows users to create markers for colonies that the model failed to detect (False Negatives). On the other hand, the 'Eraser Mode' is designed to remove incorrect detections (False Positives), such as bubbles or noise on the agar plate. An example of user interface can be shown as Fig. 4.

V. Results

To measure the performance of our developed colony detection system, we compared the results against two standard data sources. For the first model, YOLO(1), we used the original AGAR dataset, test set, for comparison. For the second model, YOLO(2) combined with k-means, the results were tested against ground truth labels verified by experts. Our evaluation focuses on two main areas, including “how accurately the system detects colonies” and “the precision of the final count”.

Regarding detection performance, outcomes were grouped into True Positives, False Positives (ghosts), and False Negatives (misses) using an IoU score of at least 0.5. Using these figures, we calculated the Accuracy, Precision, Recall, and F1-Score. These metrics provide a clear view of how successfully the model identifies and finds the exact positions of the colonies on the plates.

A. Direct Species Counting Mode

The result of Direct Species Counting mode (YOLO(1)) can be represented in TABLE I. The system achieved a high Precision of 1.000 in the detection and counting task, indicating a zero false-positive rate (no ghost detections), while maintaining an F1-Score of 0.955.

The examples of the Direct Species Counting mode (YOLO(1))’s result can be shown on Fig. 5.

TABLE I. Performance Metrics of the Direct Species Counting Mode (YOLO(1))

Task	*Acc*	*Precision*	*Recall*	*F1-score*
Detection	0.928	1.000	0.926	0.955
Species Classification	0.969	0.969	0.969	0.969

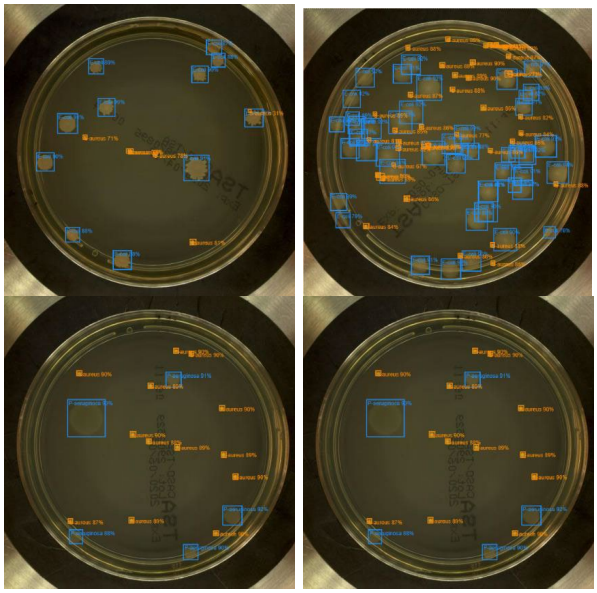


Fig. 5. Examples of the Direct Species Counting mode (YOLO(1))'s result.

B. *Selective Color-wise Counting Mode*

The result of Selective Color-wise Counting Mode (YOLO(2)) can be represented in TABLE II.

TABLE II. PERFORMANCE METRICS OF THE SELECTIVE COLOR-WISE COUNTING MODE (YOLO(2))

Task	*Acc*	*Precision*	*Recall*	*F1-score*
Detection	0.878	1.000	0.878	0.935
Color Clustering result	0.891	0.969	0.891	0.936

To evaluate the performance of the unsupervised K-means clustering, the results were subjected to a rigorous expert audit. A microbiology expert reviewed each cluster to verify whether the assigned colonies belonged to the same phenotypic group. By using the expert's assessment as the ground truth, we were able to calculate classification-style metrics (Precision, Recall, and F1-Score) for the clustering output, ensuring that the clusters reflect true biological consistency. The slight variance in clustering performance (0.891) is primarily attributed to lighting and shadow inconsistencies in certain images, which affect color value estimation. It is important to note that the performance of YOLO(1) and YOLO(2) should not be directly compared, as they operate under fundamentally different constraints. They both also trained and tested using different datasets.



Fig. 6. Examples of the Selective Color-wise Counting Mode (YOLO(2))'s result.

The examples of the Selective Color-wise Counting Mode (YOLO(2))'s result can be shown on Fig. 6.

VI. CONCLUSION AND DISCUSSION

The results show that our system works well for counting microbial colonies using two different modes. Direct Species Counting Mode (YOLO(1)) achieved a high F1-score of 0.955. This is because YOLO(1) was trained on the AGAR dataset, which has many high-quality images. On the other hand, Selective Color-wise Counting Mode (YOLO(2)) was created to be more flexible. It can count new types of bacteria that the system has never seen before by using color groups. Although YOLO(2) has slightly lower scores than YOLO(1), we should not compare them directly. They do different jobs: YOLO(1) identifies known species, while YOLO(2) adapts to new ones.

A key success of this study is that both modes reached a Precision of 1.000. This means the system is very reliable and does not count bubbles or dust as colonies (no "ghost" detections). In the Selective Color-wise mode, the Accuracy was 0.878, which is slightly lower because of shadows and lighting in some photos. However, using the HSV color space helped a lot. Unlike RGB, HSV allows the system to change or group colors without losing the shape and details of the colonies.

However, we must admit that our system is not perfect. Many factors can sometimes cause the system to miss some colonies. Because of these limitations, our work emphasizes a Human-in-the-loop approach. By allowing experts to verify and correct the model's results, we combine the speed of technology with the high accuracy of human judgment.

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